Z+T Ch 4 Taylor Series ~~+ series sol'ns~~ 9-16

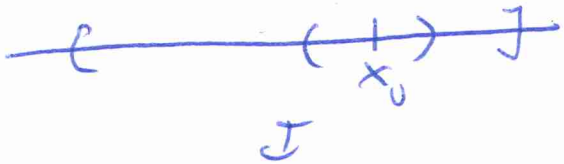
How to find Taylor series in multiple dimensions

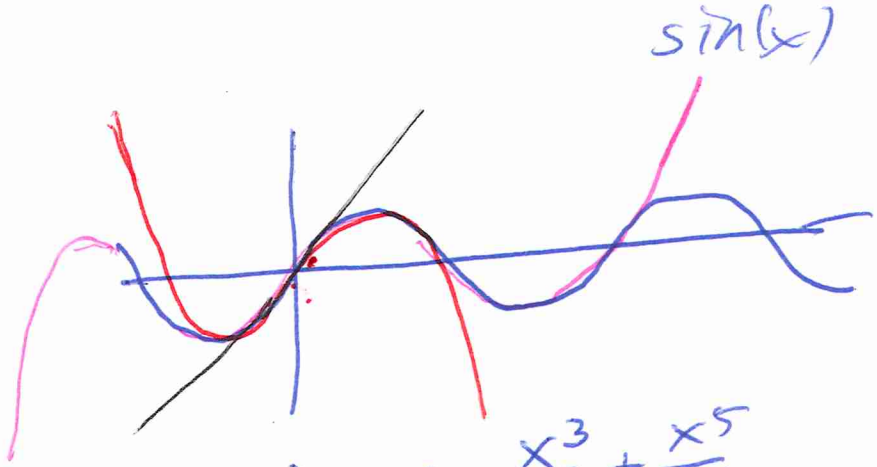
<u>Taylor series in one dim</u>

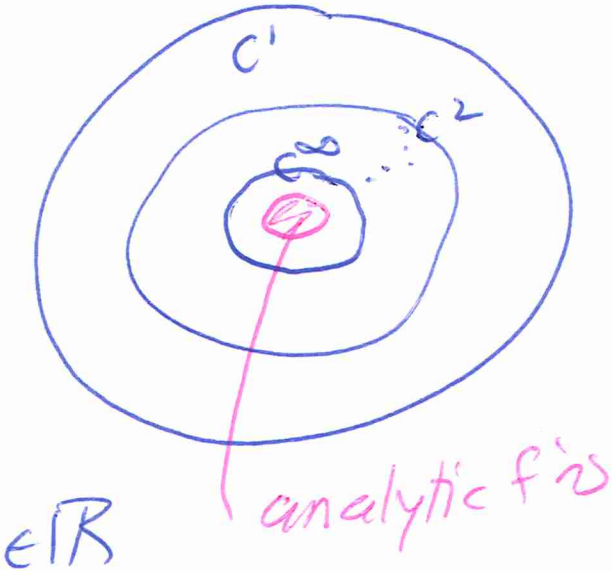
$$f(x) = f(x_0) + f'(x)(x-x_0) + \frac{1}{2!} f''(x)(x-x_0)^2 + \dots$$

~~If~~ $f \in C^\infty$, Then Taylor series for $f(x)$ exists, although the ∞-series may or may not converge, or if it does converge, it may not converge to $f(x)$ (problem 1.1, on HW).

<u>Def'n</u> Given $x_0 \in I$, $f \in C^\infty(I)$, Then f is said to be <u>analytic</u> at x_0 if The Taylor series about x_0 converges to $f(x)$ in a nbd of x_0.







$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

∞-terms, conv. for all $x \in \mathbb{R}$

Taylor series in Multidimensions

Notation $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$ symbol (not a vector) for $n=2$ in $\mathbb{R}^2$, $n=3$ in $\mathbb{R}^3$, ...

$$D^\alpha f = D_1^{\alpha_1} D_2^{\alpha_2} \cdots D_n^{\alpha_n}$$

deriv. wrt 1st indep

wrt 2nd indep

wrt nth indep.

how many deriv.

Also: $x^{\alpha} \triangleq x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}$ $\triangleq$ def'n

$\alpha! \triangleq \alpha_1! \, \alpha_2! \cdots \alpha_n!$ $\alpha_i \in \mathbb{N}$ $1, 2, 3, \ldots$

$|\alpha| \triangleq \alpha_1 + \alpha_2 + \cdots + \alpha_n$

$$\boxed{\text{Taylor series of } f(\underline{x}) \text{ about } \underline{x} = \underline{x}^0 \quad \sum_{|\alpha| \geq 0} \frac{D^{\alpha} f(\underline{x}^0)}{\alpha!} (\underline{x} - \underline{x}^0)^{\alpha}}$$

<u>Def'n</u> Given $f \in C^{\infty}(\Omega)$, $\Omega \subset \mathbb{R}^n$, $\underline{x}^0 \in \Omega$, then f is <u>analytic</u> at $\underline{x}^0$ if the Taylor series converges to $f(\underline{x})$ for all $\underline{x}$ in some ~~nd~~ nbd of $\underline{x}^0$.

ex) 1.5c) $f(x,y) = x\ln(1+x+y) - y^2$

Find Taylor series up to order 2 (i.e. $0 \le |\alpha| \le 2$) about $(0,0)$.

$\alpha = (0,0), (1,0), (0,1), (1,1), (2,0), (0,2)$

for $|\alpha| = 3$: $(3,0), (0,3)$
$(2,1), (1,2)$

Taylor series: $\frac{1}{0!\,0!} D^{(0,0)} f(0,0)\,(x-0)^0 (y-0)^0$

$+ \frac{1}{1!\,0!} D^{(1,0)} f(0,0)\,(x-0)^1 (y-0)^0$

$+ \frac{1}{0!\,1!} D^{(0,1)} f(0,0)\; y$

$+ \frac{1}{1!\,1!} D^{(1,1)} f(0,0)\; x y$

$+ \frac{1}{2!\,0!} D^{(2,0)} f(0,0)\, x^2 + \frac{1}{0!\,2!} D^{(0,2)} f(0,0)\, y^2$

$\alpha = (0,0)$

$$f(x,y) = x\ln(1+x+y) - y^2 \qquad f(0,0) = 0$$

$(1,0)$ $$\frac{\partial f}{\partial x}(x,y) = \ln(1+x+y) + \frac{x}{1+x+y} \qquad \frac{\partial f}{\partial x}(0,0) = 0$$

$(0,1)$ $$\frac{\partial f}{\partial y}(x,y) = \frac{x}{1+x+y} - 2y \qquad \frac{\partial f}{\partial y}(0,0) = 0$$

$(2,0)$ $$\frac{\partial^2 f}{\partial x^2}(x,y) = \frac{(1+x+y) - x(1)}{(1+x+y)^2} + \frac{1}{1+x+y} \qquad \frac{\partial^2 f}{\partial x^2}(0,0) = 2$$

$(1,1)$ $$\frac{\partial^2 f}{\partial x \partial y}(x,y) = \frac{1(1+x+y) - x(1)}{(1+x+y)^2} \qquad \frac{\partial^2 f}{\partial x \partial y}(0,0) = 1$$

$(0,2)$ $$\frac{\partial^2 f}{\partial y^2}(x,y) = \frac{-x}{(1+x+y)^2} - 2 \qquad \frac{\partial^2 f}{\partial y^2}(0,0) = -2$$

$$1(0) + 1(0) + (1)(0) + (1)(1)xy$$
$$+ \frac{1}{2}(2)x^2 + \frac{1}{2}(-2)y^2 = \boxed{xy + x^2 - y^2}$$

Thm (Cauchy-Kovalevsky Thm) not responsible for

Consider the Cauchy Problem

$$\frac{\partial u}{\partial t} = F(t, x_1, x_2, \ldots, x_n, u, u_{x_1}, u_{x_2}, \ldots, u_{x_n})$$

1st-order PDE, one dependent variable & n+1 indep. variables

$$u(0, x_1, \ldots, x_n) = \phi(x_1, \ldots, x_n)$$

If ϕ is analytic about the origin (think combinations of polynomials, exp, sin, cos) and F is analytic about initial curve in $\mathbb{R}^{2n+2}$, then there exists a ~~unique~~ sol'n which is analytic and unique in the class of analytic f'ns.

$f(x,y) = 1 - x^3 + y^2$ for any (x^0, y^0)

Taylor series $|\alpha| \leq 4$: $1 - x^3 + y^2$

$\boxed{\text{All polynomials are analytic!}}$